

Alexander Cai

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EDUCATION

Harvard University
Ph.D. Computer Science

Cambridge, MA
Sep 2025–Current

- Advised by Prof. Kianté Brantley
- Awarded Harvard Kempner Institute Graduate Fellowship

Harvard University
S.M. Applied Mathematics

Cambridge, MA
Sep 2021–May 2025

A.B. Statistics and Computer Science (cum laude), Secondary in German Studies (concurrent)

- **Relevant coursework:** Foundations of Deep Learning, Topics in High-Dimensional Statistics, Neural Computation, Computing at Scale, Physical Mathematics, Operating Systems, Distributed Systems

EXPERIENCE

Brantley Lab (Harvard University)
Graduate Research Assistant

Cambridge, MA
Sep 2025–Current

- Design imitation learning algorithms that generalize from few samples

Harvard University
Head Teaching Fellow, CS 1840: Introduction to Reinforcement Learning

Cambridge, MA
Sep 2026–Current

- Lead course staff for Prof. Kianté Brantley’s course: TF/CA hiring, course compute environment, exams, and an Embedded EthiCS module on RLHF

Gershman Lab (Harvard University)
Research Intern

Cambridge, MA
Jan–Aug 2025

- Implement a multitask RL algorithm based on the “multitask preplay” model from cognitive science
- Run distributed RL experiments with Weights & Biases on a high-performance compute cluster

Harvard University
Course Assistant

Cambridge, MA
Sep 2022–Dec 2024

- Led sections, designed course materials, and graded assignments for CS/Stat 184(0): Reinforcement Learning, CS 181(0): Machine Learning, AM 226: Theory of Neural Computation
- Authored course textbook for CS/Stat 184(0) also used for Cornell’s CS 4789

QuantCo
Software Engineering Intern

Berlin, Germany
Jun 2024–Aug 2024

- Designed a scalable Python framework for feature engineering

Pehlevan Group (Harvard University)
Research Intern

Cambridge, MA
May 2023–Aug 2023

- Implemented a novel unsupervised algorithm for 2D-to-3D video tracking based on transport operators
- Awarded Program for Research in Science and Engineering (PRISE) Fellowship

- Accelerated terabyte-scale bulk import tasks by up to 5x in DataStax’s Cloud Native Database
- Developed the **sstable-to-arrow** utility for reading Apache Cassandra database files into the fast Apache Arrow memory format

PUBLICATIONS

- D. Rooryck*, **A. Cai***, Yonatan Belinkov, David Alvarez-Melis, Kianté Brantley. “Feeding BabyLMs Macaroni: Code-Switching Curricula Cause Cross-Lingual Convergence,” 2026. EMNLP 2026 BabyLM Workshop. <https://arxiv.org/abs/2609.30535>
- J. Roggeveen*, E. Wang* et al. (incl **A. Cai**). “HARDMath2: A Benchmark for Applied Mathematics Built by Students as Part of a Graduate Class,” 2025. NeurIPS 2025. <https://arxiv.org/abs/2505.11774>
- **A. Cai**. “An Introduction to Reinforcement Learning,” 2025. Undergraduate thesis. <https://rlbook.adzc.ai>
- L. Tang, T. Shlomi, **A. Cai**. “Learning the Wrong Lessons: Inserting Trojans During Knowledge Distillation,” 2023. ICLR 2023 BANDS Workshop. <https://arxiv.org/abs/2303.05593>
- L. Tang, **A. Cai**, S. Li, J. Wang. “The Naughtyformer: A Transformer Understands Offensive Humor,” 2022. AAAI-23 Student Abstract. <https://arxiv.org/abs/2211.14369>
- W. Luo, R. Chin, **A. Cai**, G. Lin, B. Su, A. Zhang. “A tardiness-augmented approximation scheme for rejection-allowed multiprocessor rescheduling,” 2022. Journal of Combinatorial Optimization 44, 690–722. <https://link.springer.com/article/10.1007/s10878-022-00857-y>
- **A. Cai**. “Analyzing Cassandra Data Using GPUs,” 2021. <https://developer.nvidia.com/blog/analyzing-cassandra-data-using-gpus-part-1/>

TALKS

- Kempner Lunch and Learn. Cross-lingual transfer in multilingual models. 8 October 2026.
- Kempner Lunch and Learn. Maximum entropy reinforcement learning. 16 September 2025.
- Kempner Lunch and Learn. A brief history of perspectives in language acquisition. 16 July 2025.

POSTERS

- Kempner Institute Workshop: Learning Dynamics in Natural and Artificial Intelligence. Investigating code-switched curriculum learning in data-constrained multilingual modelling. 2 September 2026.
- Kempner Institute Spring Into Science. Grammar acquisition trajectories of developmentally plausible bilingual models. 21 May 2026.

ACTIVITIES AND LEADERSHIP

LGBTQ@GSAS, a Harvard Griffin GSAS Student Group

Vice President

Cambridge, MA

Sep 2026–Current

- Organize community-building events for the queer graduate student community.

Derek Bok Center for Teaching and Learning

Teaching Development Fellow

Cambridge, MA

Jun–Sep 2026

- Train undergraduate course assistants on teaching responsibilities and pedagogy

InTouch (Harvard SEAS)*Peer Staffer*

Cambridge, MA

May 2026–Current

- Organize community building for InTouch, a peer-to-peer support network for SEAS graduate students

Currier House, Harvard College*Non-Resident Tutor*

Cambridge, MA

Sep 2025–Current

- Mentor undergraduates and support residential community life as an affiliated graduate tutor

AddisCoder*Teaching Assistant*

Addis Ababa, Ethiopia

Aug 2025, Aug 2026

- Teach computer science fundamentals to 120 high school students

The Harvard Crimson*Director of Graphics*

Cambridge, MA

Mar 2024–Dec 2024

- Designed interactive visualizations to illustrate student newspaper articles
- Directed team of 10 undergraduate designers and programmers
- Finalist for the Society of Professional Journalists Mark of Excellence Award

Harvard Undergraduate Machine Intelligence Community (HUMIC)*Co-Director of Fellowships*

Cambridge, MA

Apr 2022–Dec 2023

- Design and teach fellowship program about modern ML and AI to 70 undergraduates
- Lead weekly paper readings on state-of-the-art developments in ML and AI

Harvard College Project for Asian and International Relations (HPAIR)*Co-Director of Technology*

Cambridge, MA

Oct 2021–Aug 2023

- Designed and built application portal at my.hpair.org, enabling over 20,000 applications
- Integrate Firebase with BigQuery and Google Sheets to provide a transparent and accessible view

SKILLS AND INTERESTS

Programming: Jax RL ecosystem, PyTorch, C++, Apache Arrow, TypeScript, Firebase, Redis, SQL

Languages: English (native), Mandarin (HSK 6), German (B2), French (B2)

Awards: Germanic Languages and Literatures Book Prize, CyberTitan III National Silver Medalist, University of Toronto National Book Award, Canadian Open Mathematics Challenge Gold Award, Alberta Winter Games Men's Sabre Gold Medalist

Interests: Reinforcement learning, linguistics, web development, historical fiction, trombone, musical theater